Field Hours

CSCE 743 Digital Fabrication Studio

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

Beam on, beam off

You describe a machine by the path a tool follows and by what it does to material. The laser travels in x and y over a bed of 400 by 300 millimeters, and it cuts only where the beam is on. A segment that leaves the bed is a fault, not a cut. You write segments with a beam state, on or off, rather than a picture of the part. The picture is not yet a toolpath.

You separate drawing length from cutting time. A 40 millimeter square becomes four beam-on sides, 160 millimeters of cutting, plus beam-off travel between jobs. At a feed of 20 millimeters per second, those 160 millimeters take 8 seconds of beam-on time. You compute that before you start the machine. Beam-off moves add time, but they do not add char. The mechanism is the state on each segment, not the outline in the file.

A drawing becomes segments. Time comes from beam-on length and feed.

Drawing on the bed -> Segments, beam state -> Feed of 20 mm/s -> Time from the length

A 40 millimeter square

1. The bed is 400 by 300 millimeters. Each side of the square is 40 millimeters, so the square fits.

2. Beam-on length is 4 times 40, which is 160 millimeters.

3. At 20 millimeters per second the beam is on for 8 seconds. Travel with the beam off is extra time.

Problem 1. Four sides of 40 millimeters are cut at 20 millimeters per second. How many seconds is the beam on?

Answer. 8 seconds. The length is 160 millimeters, and 160 divided by 20 is 8.

Problem 2. A segment runs past the 400 millimeter edge of the bed. What is it?

Answer. A fault, not a cut. The toolpath has to stay on the bed. You do not send it because the drawing looked small on the screen.

Lecture 2 · 90 minutes

Plywood, kerf, and the wrong knife

You choose a material by how it fails. This sheet is 3 millimeter plywood. The laser chars an edge. The vinyl cutter will not cut it: the sheet is too thick for a drag knife. You write 3 millimeters on the job. A path made for 6 millimeter stock does not transfer. On a laser it may burn through, because the pass count was chosen for the thicker stock.

You account for kerf, the width the tool destroys. This laser kerf is 0.2 millimeters. A slot drawn at 10

millimeters becomes about 10.2 millimeters if the beam walks the center of the line. You offset the path by half the kerf, 0.1 millimeters, into the waste, so the part that remains matches the drawing. You cut and measure one test square before the rest of the sheet. The charred band is not solid material you can count on.

The drawing is the intent. The kerf offset is what the part actually keeps.

Drawn slot

Slot drawn at 10 mm

Path on the line

No offset yet

After the kerf

Kerf is 0.2 mm

Offset by 0.1 mm

Offset goes into waste

A slot in 3 millimeter plywood

1. The sheet is 3 millimeter plywood. The laser kerf is 0.2 millimeters. The vinyl knife is the wrong tool.

2. You offset each side of the 10 millimeter slot by 0.1 millimeters into the waste.

3. You cut one test square and measure it before you commit the rest of the sheet.

Problem 1. The kerf is 0.2 millimeters and you offset into the waste. How far does the path move?

Answer. 0.1 millimeters, half the kerf. Offsetting by the full 0.2 would remove too much of the part.

Problem 2. Why do you refuse the vinyl cutter for this sheet?

Answer. The sheet is 3 millimeter plywood, too thick and too hard for a drag knife. The laser matches this material. The vinyl cutter is for thin film.

Lecture 3 · 90 minutes

Packing the bed

You pack parts on the 400 by 300 millimeter bed with a gap so neighboring kerfs do not eat one part. Each gap is at least 2 millimeters. Five squares of 40 millimeters, plus 4 gaps of 2, occupy 208 millimeters. A 10 millimeter margin leaves a usable run of 380. The row of 208 fits. You do not shave a gap to 0 to squeeze another square.

You order cuts so a freed part cannot shift. Cut interior holes before the outline. If the outline of a 40 millimeter square runs first, the square can tilt and a later hole will miss. This job has 1 hole and 1 outline, so the hole runs first. You also leave a 1 millimeter tab so the part does not fall into the bed. You cut that tab by hand after the sheet comes off.

Pack with a gap, cut holes before outlines, and hold loose parts with a tab.

1. Gap parts by 2 mm

2. Holes before outlines

3. Leave a 1 mm tab

4. Cut the tab by hand

Five squares on the bed

1. Each square is 40 millimeters, and each of the 4 gaps is 2 millimeters. The row is 208 millimeters.

2. The bed is 400 millimeters wide, with a 10 millimeter margin, so 208 fits inside the usable 380.

3. Each hole is cut before its outline, and a 1 millimeter tab holds the part until the sheet is off the machine.

Problem 1. Five squares of 40 millimeters have 4 gaps of 2 millimeters. How long is the row?

Answer. 208 millimeters. Five times 40 is 200, and four times 2 is 8.

Problem 2. A part has one hole and one outline. Which path runs first, and what holds the part?

Answer. The hole runs first. If the outline runs first, the part can shift and the hole misses. A 1 millimeter tab can hold the part, and you cut that tab after the sheet leaves the machine.

Lecture 4 · 90 minutes

Fifty layers

You print by stacking layers, and each layer is its own toolpath. Layer height is 0.2 millimeters. A part 10 millimeters tall needs 50 layers. Each layer has an outline plus infill. You do not treat the part as one laser-style path. A fault in layer 12 is buried by layer 30. You cannot recut it the way you can recut an open sheet.

You set infill as a fraction of the layer area, in the job, not by drawing interior walls by hand. At 20 percent the interior is mostly air, and the wall still follows the outline. The same 10 millimeter cube at 80 percent uses more material and compresses less. The nozzle diameter is 0.4 millimeters, so one wall line is about that wide. Those walls have to sit inside the outer size, or the cube prints larger than 10 millimeters.

A print is a stack of toolpaths. A bad lower layer is covered before you can see it fail.

50 layers at 0.2 mm

Nozzle width 0.4 mm

Infill at 20 percent

A lower fault is buried

A 10 millimeter cube

1. Height 10 millimeters at 0.2 millimeters per layer is 50 layers.

2. The nozzle is 0.4 millimeters wide, and the walls consume that width inside the outer size.

3. Infill is 20 percent of each layer. The same cube at 80 percent uses more material and compresses less.

Problem 1. The part is 10 millimeters tall and each layer is 0.2 millimeters. How many layers do you print?

Answer. 50. You divide 10 by 0.2.

Problem 2. Why is a bad layer 12 hard to repair once you are on layer 30?

Answer. Layer 12 is already covered. The print is 50 stacked toolpaths, not one path you can stop and recut the way you might on an open laser sheet.

Lecture 5 · 90 minutes

A bit that cannot cut a sharp corner

You mill by removing stock with a spinning bit, and the diameter limits the inside corner. A 6 millimeter end mill cannot cut an inside corner sharper than radius 3, half the diameter. A square pocket comes off with rounded corners. You redesign the corner to a radius of at least 3 millimeters, or you plan a second operation. The bit will not make a radius of 0.

You leave stock for a finish pass. The rough pass stays 0.5 millimeters proud of the final surface, and the finish pass removes that 0.5. If you rough to final size, a deflection gouge cannot be repaired. The block is larger than the part: you leave 20 millimeters of extra stock so the bit never meets the vise jaws.

The drawing can ask for a square corner. The bit can only leave its own radius.

Drawn pocket

Inside radius asked: 0

Final size already

No stock left to fix

6 mm end mill

Radius at least 3 mm

Rough leave of 0.5 mm

Finish removes that 0.5

A pocket and a 6 millimeter bit

1. The bit diameter is 6 millimeters, so the inside corner radius cannot be under 3 millimeters.

2. The rough pass stops 0.5 millimeters before the final surface.

3. The finish pass removes that 0.5. The vise holds an extra 20 millimeters of stock so the bit stays off the jaws.

Problem 1. The end mill diameter is 6 millimeters. What is the smallest inside corner radius it can cut?

Answer. 3 millimeters, half the diameter. A drawn corner of radius 0 will not appear on the part.

Problem 2. Why does the rough pass stop 0.5 millimeters early?

Answer. So the finish pass still has stock to remove. If you rough to the final size, a deflection gouge cannot be corrected because the material is already gone.

Lecture 6 · 90 minutes

Eight tiles and a rule

You treat programmable matter as parts that carry an assembly rule, not as a printer setting. This studio uses 8 identical tiles. Each has a west edge and an east edge that fit in only 1 orientation. A tile checks its neighbor, not a drawing of the whole chain. You build the 8 by those edges, and you stop if a joint seats only when you flip a tile against its mark.

You make the rule physical in the joint. A tab 10 millimeters wide, cut so that the 0.2 millimeter kerf does not loosen it, seats one way. You offset by 0.1 millimeters, half the kerf, as on the plywood sheet. A symmetric tab seats two ways, and then the neighbor cannot tell the legal orientation. You cut the 8 tiles from the 3 millimeter plywood. The machine does not assemble them. You do. The programmed part is the joint.

The joint is the program. The laser only cuts it. You assemble the chain.

1. Eight identical tiles

2. West edge, east edge

3. The joint fits one way

4. Stop if a tile flips

A chain of eight

1. Each tile west edge accepts only the previous tile east edge.

2. The tab is 10 millimeters. A 0.1 millimeter offset keeps the 0.2 millimeter kerf from opening a second orientation.

3. You seat all 8 by hand. If tile 5 fits only when flipped, the mark is wrong and you stop.

Problem 1. A tab is symmetric and seats in 2 orientations. What happens to the assembly rule?

Answer. The local rule is lost. A neighbor can no longer read the legal orientation from the joint. You need an asymmetric tab, such as the 10 millimeter one-way joint.

Problem 2. Who assembles the 8 tiles?

Answer. You do, by hand. The laser only cuts them from the 3 millimeter sheet. The programmed behavior is in the joint, not in a machine step that snaps the chain together.

Lecture 7 · 90 minutes

Two tiles before eight

You test one joint before cutting 8 tiles. Cut 2 tiles, seat them, and pull. If they fall apart, the 0.1 millimeter offset was too large, or the sheet is under the 3 millimeters you assumed. If they will not seat, the offset was too small. You change the offset by 0.05 millimeters and cut 2 more. You do not recut all 8 from one sticky joint.

You record the test the way a studio records a critique you can reuse. Write the machine, the thickness, the kerf you assumed, the offset you used, and whether the joint seated. A later note about the 10 millimeter printed cube does not teach you about this plywood joint. Different machine, different failure. When the 2-tile test seats and resists a pull, then you cut the remaining tiles toward a chain of 8.

A pair of tiles is the experiment. The batch of eight waits on that result.

Cut 2 test tiles -> Seat them and pull -> Adjust by 0.05 mm -> Then cut the rest

A loose pair

1. Two tiles in 3 millimeter plywood fall apart. The offset you used was 0.1 millimeters.

2. You reduce the offset by 0.05, to 0.05 millimeters, and you cut two new tiles.

3. They seat and hold under a pull. Only then do you cut the tiles that complete a chain of 8.

Problem 1. Two test tiles fall apart and the offset was 0.1 millimeters. What do you cut next?

Answer. Two new tiles at an offset reduced by 0.05 millimeters. You do not recut all 8 until a pair seats and holds a pull.

Problem 2. Why keep the laser notes off the page that records the 10 millimeter printed cube?

Answer. They are different machines and different failures. The cube layer height does not say whether the plywood kerf offset was right. The joint test stays with the laser job.

Lecture 8 · 90 minutes

A job sheet you can trust

You can now turn a drawing into a toolpath on a 400 by 300 millimeter bed, offset a 0.2 millimeter kerf, pack parts with 2 millimeter gaps, print 50 layers of 0.2 millimeters, and leave 0.5 millimeters where a mill cannot beat an inside radius of 3. You can cut 8 tiles that assemble one way, after a 2-tile pull test. The skill is the path, the material, and a test you can hold.

You finish a piece only when the test and the batch agree. A cube of 50 layers at 20 percent infill is not the plywood chain, and sanding one hides nothing about the other. Before 8 tiles you need the 2-tile pull test. Before a pocket you need the corner a 6 millimeter bit can cut. You write those limits on the job sheet and stop when a path would cross one.

Each machine has its own limit. A pass on one does not clear the other.

Laser and plywood

Kerf of 0.2 mm

Test 2 tiles first

Then a chain of 8

Mill and printer

Corner radius at least 3

Finish leave of 0.5 mm

50 layers of 0.2 mm

Three jobs, three results

1. The laser job names 3 millimeter plywood, a starting offset of 0.1 millimeters, and a 2-tile test before 8 tiles.

2. The mill job names a 6 millimeter bit, a minimum inside radius of 3 millimeters, and a rough leave of 0.5 millimeters.

3. The print job names 50 layers and 20 percent infill. You do not borrow a pass or a fail from one job for another.

Problem 1. Combine kerf and the assembly rule. The kerf is 0.2 millimeters, and a symmetric tab seats two ways. What do you change, and what offset do you start from?

Answer. Make the tab asymmetric so only one orientation seats, and offset the path by half the kerf, 0.1 millimeters, into the waste. Prove the joint on 2 tiles before you cut 8.

Problem 2. Combine the bed limit and the mill. The drawing asks for a square inside corner, and the only bit has diameter 6 millimeters. The blank must also stay on the 400 by 300 millimeter bed. What radius can you cut?

Answer. 3 millimeters, not 0. The bit cannot make a sharper inside corner, so the drawing changes or you plan a second operation. Separately, the toolpath still has to stay inside the 400 by 300 millimeter bed.